

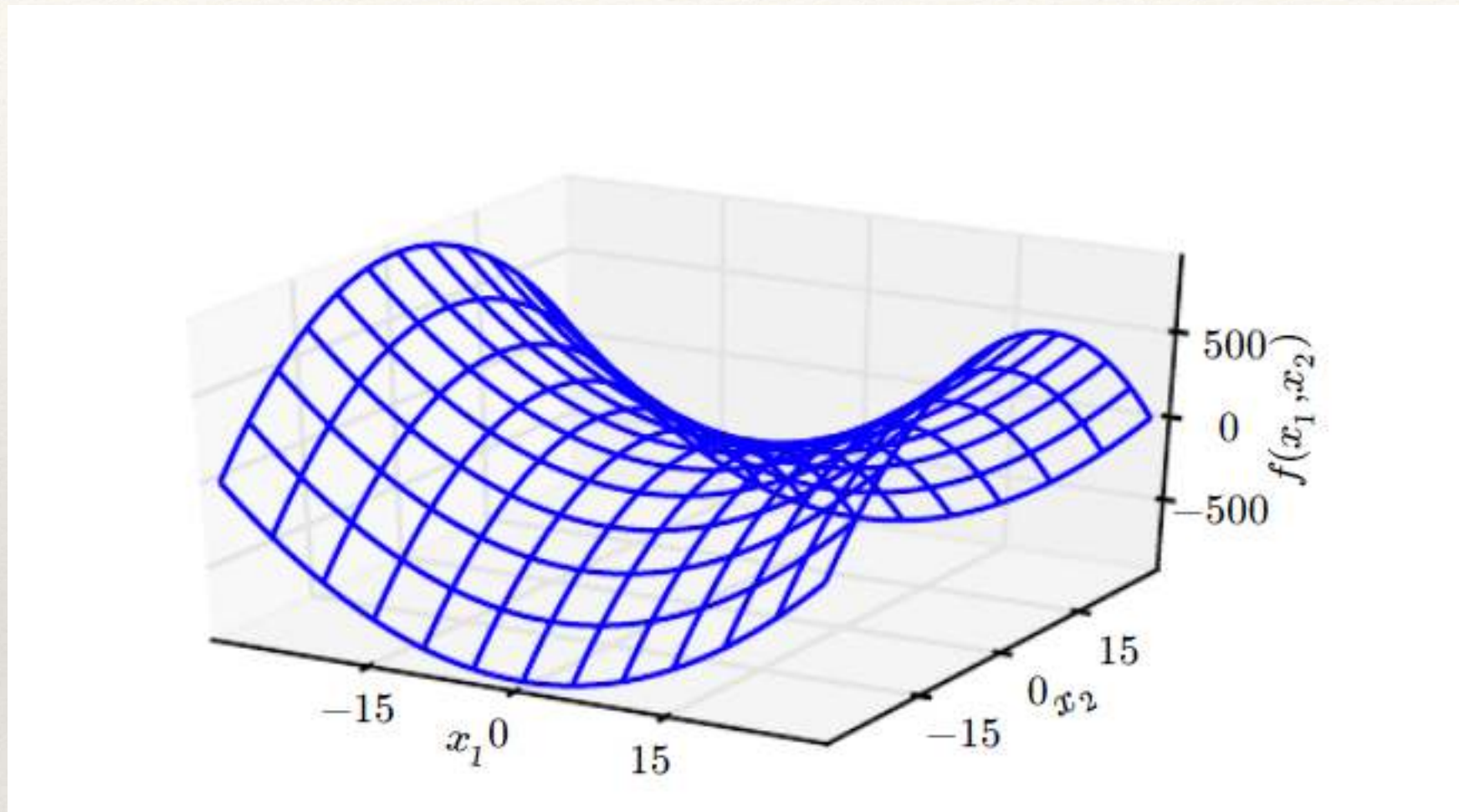
# *E9 205 Machine Learning for Signal Processing*

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**Neural Networks - Learning, Error  
Functions and Properties**

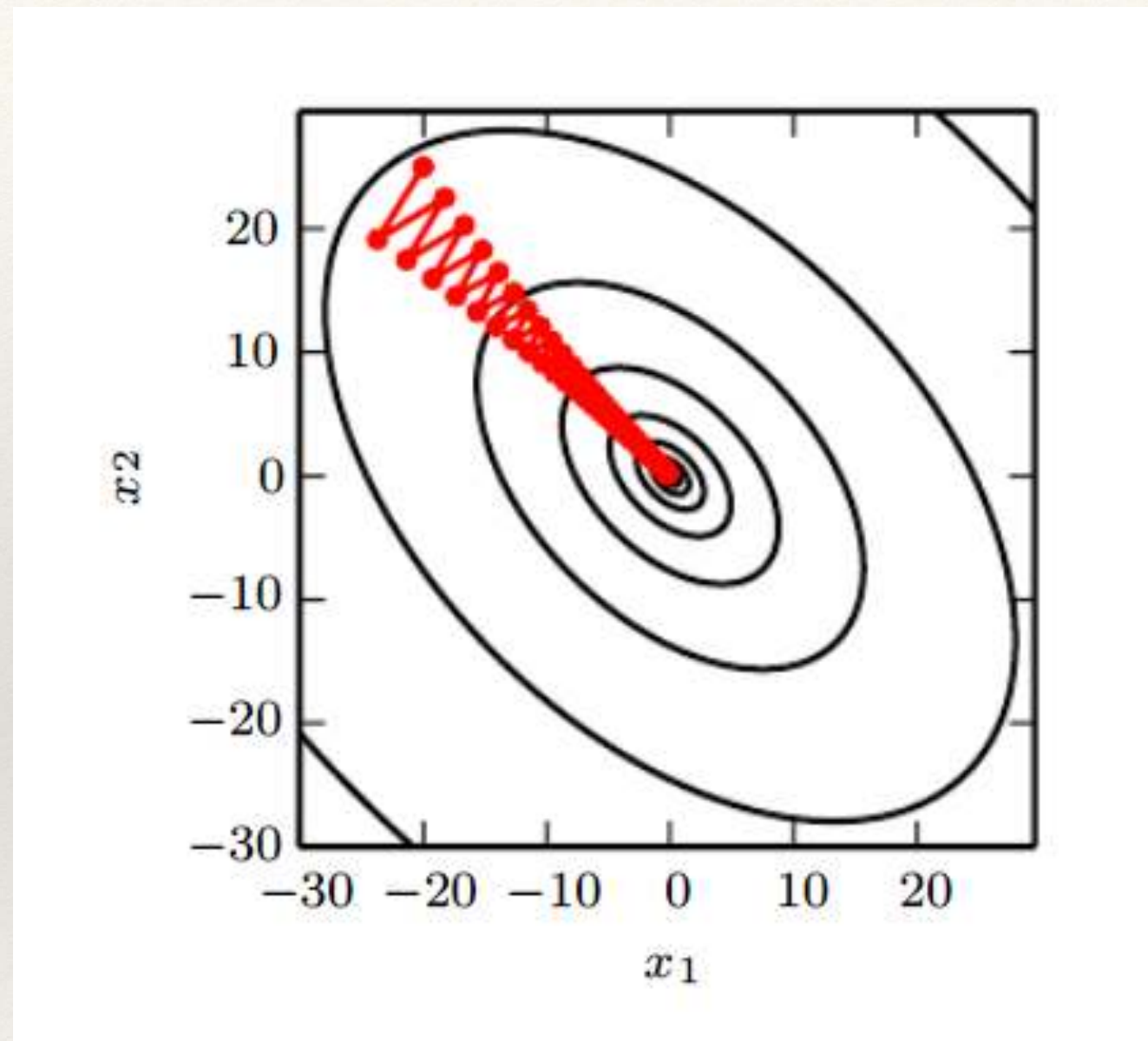
13-11-2017

# Directional Derivatives (Max and Min)



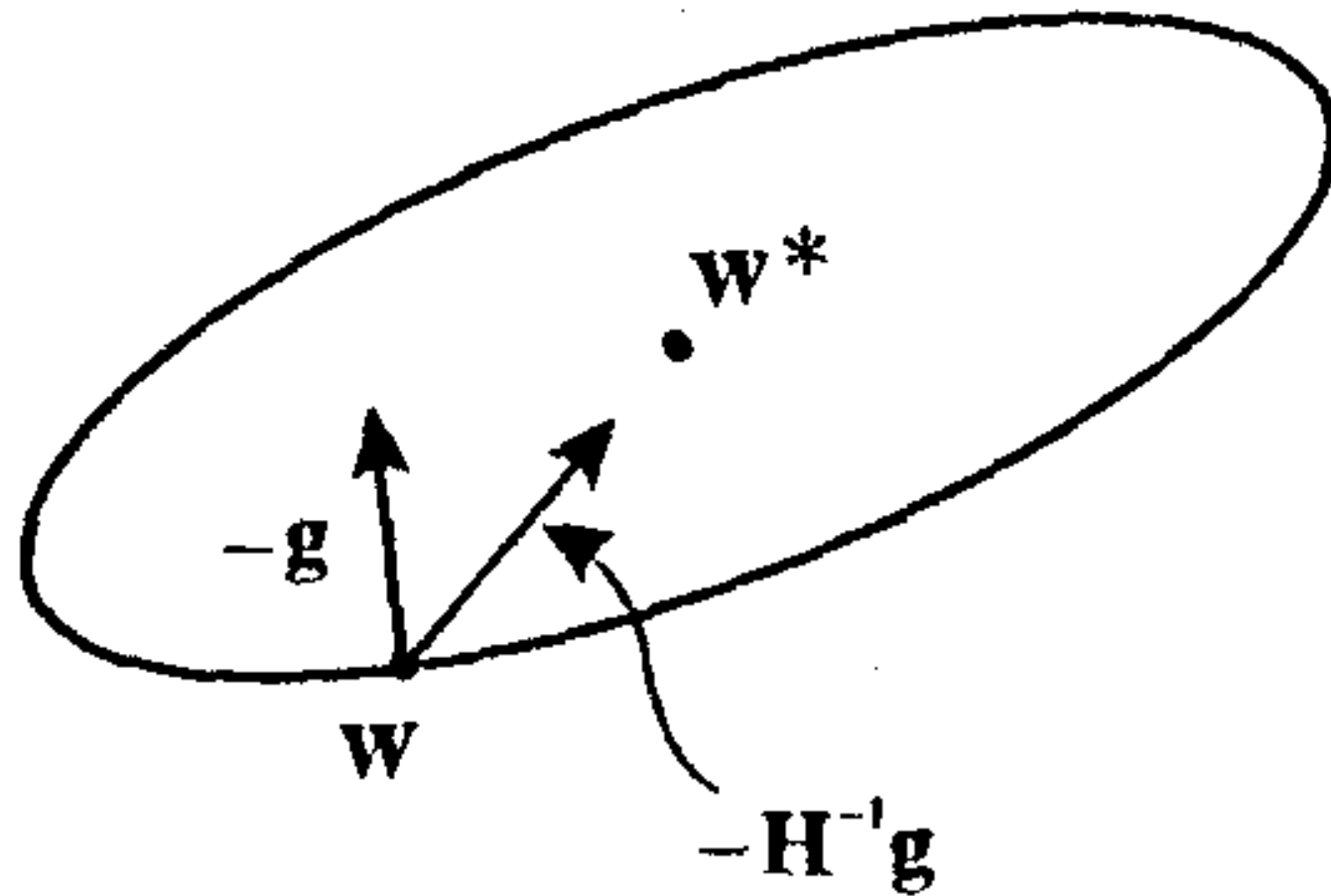
$$f(\mathbf{x}) \approx f(\mathbf{x}^{(0)}) + (\mathbf{x} - \mathbf{x}^{(0)})^\top \mathbf{g} + \frac{1}{2}(\mathbf{x} - \mathbf{x}^{(0)})^\top \mathbf{H}(\mathbf{x} - \mathbf{x}^{(0)}).$$

# Issues with Gradient Descent



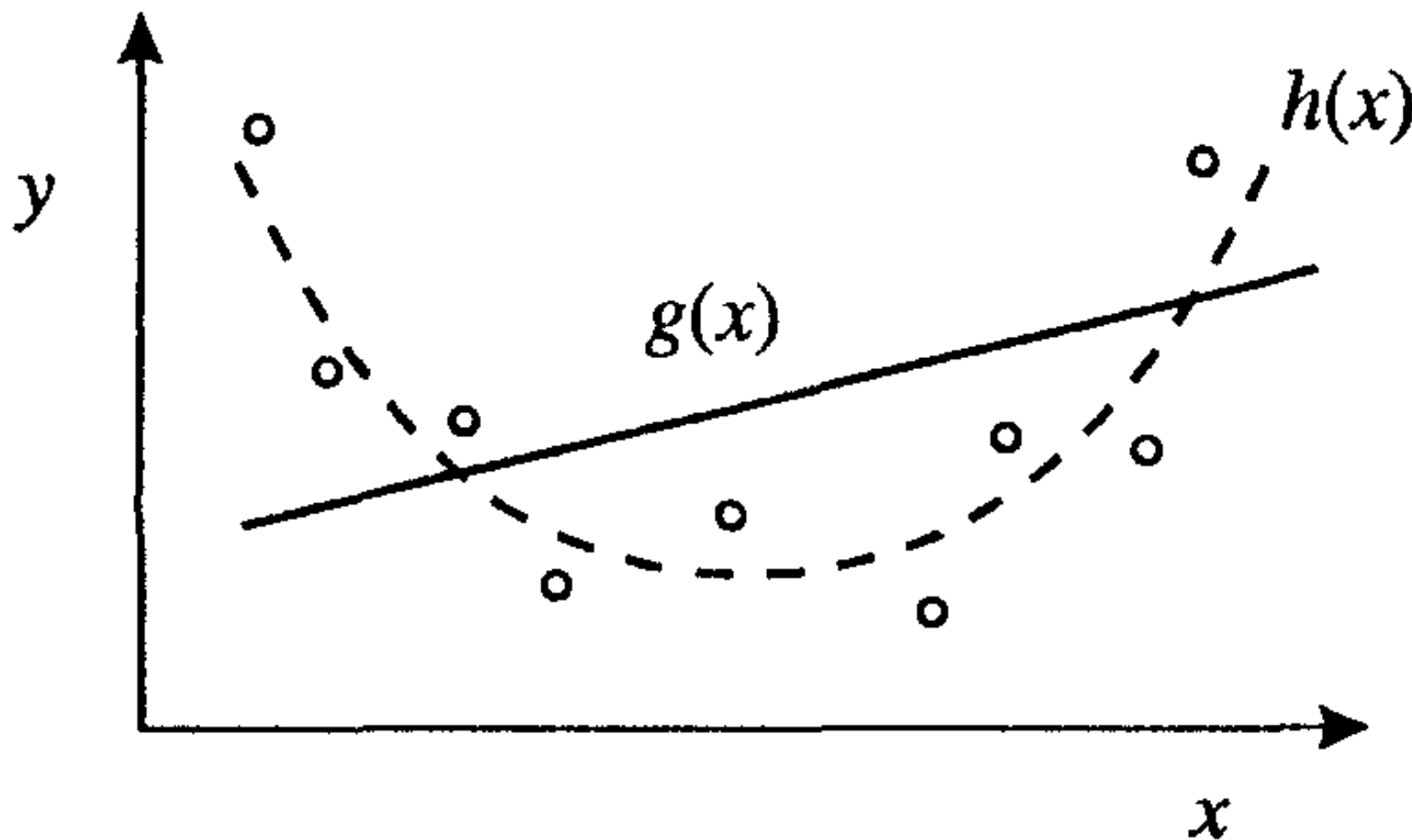
$$f(\mathbf{x}) \approx f(\mathbf{x}^{(0)}) + (\mathbf{x} - \mathbf{x}^{(0)})^\top \mathbf{g} + \frac{1}{2}(\mathbf{x} - \mathbf{x}^{(0)})^\top \mathbf{H}(\mathbf{x} - \mathbf{x}^{(0)}).$$

# Issues with Gradient Descent



$$f(x) \approx f(x^{(0)}) + (x - x^{(0)})^\top g + \frac{1}{2}(x - x^{(0)})^\top H(x - x^{(0)}).$$

# Bias Variance Trade Off



# Bias Variance Trade Off

